

Research Gap and Opportunity Report

Existing approaches, weaknesses, open questions, closest prior art, 15 ideas and 5 paper concepts

Machine Learning for Alpha Signal Combination and Portfolio Construction

This report is part of a pre-empirical research proposal. It distinguishes verified evidence, research inference, proposed design choices and hypotheses. No backtest results are reported; expected outcomes are hypotheses.

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Research Gap and Opportunity Report

Executive Summary

A research gap is recorded here only if a systematic comparison of the literature shows that a question is (i) economically important, (ii) not answered by existing papers, and (iii) testable with obtainable data. "Future work" sentences in papers are not treated as gaps. Each gap has a status (Solved / Partially solved / Open / Open but low value) and a pointer to the closest prior art. The closest-prior-art tracker (Closest_Prior_Art_Tracker.md) is a **living document**: it must be re-checked at each project phase and immediately before submission.

The central finding is that the literature has already established three things: ML improves prediction, costs destroy much of ML's gross alpha, and learning under a cost-aware economic objective restores much of it (Gu, Kelly & Xiu, 2020; Avramov, Cheng & Metzker, 2023; Jensen et al., 2026). What it has **not** done is identify *which* ingredients of ML signal combination create net-of-cost value, how those ingredients interact, and whether the resulting state dependence is economically interpretable. That decomposition question is where we see a defensible contribution [inference].

1. Existing Approaches and Their Weaknesses

Approach	Representative papers	Strength	Weakness for net investment value
Naive/heuristic combination (equal weight, IC-weight, theme-EW)	Grinold & Kahn (2000); Grinold (1989); Rapach, Strauss & Zhou (2010)	Robust, low turnover, hard to beat	Ignores interactions, redundancy, costs, states
Linear statistical combination (FM, ridge, PLS)	Lewellen (2015); Light, Maslov & Rytchkov (2017); Freyberger, Neuhierl & Weber (2020)	Transparent, stable	Misses nonlinearity; static; prediction loss
Shrunk SDF / mean-variance over signal portfolios	Kozak, Nagel & Santosh (2020)	Economically grounded	Portfolio-level; turnover at stock level not modelled
Nonlinear ML prediction then sort	Gu, Kelly & Xiu (2020); Cakici et al. (2023); Tobek & Hronec (2021)	Best statistical accuracy	Gross sorts; microcap/illiquid concentration (Avramov, Cheng & Metzker, 2023)
ML prediction then optimiser (predict-then-optimize)	BlackRock Systematic (2025); Markowitz-ML in Jensen et al. (2026)	Modular, industrial	Loss-objective mismatch; error maximisation; costs only ex post
Parametric/economic-objective weight learning	Brandt, Santa-Clara & Valkanov (2009); DeMiguel et al. (2020); Simon, Weibels & Zimmermann (2026); Jensen et al. (2026)	Directly optimises net utility	Mostly static mapping; attribution of gains unclear [inference]
End-to-end differentiable optimisation	Butler & Kwon (2023); Costa & Iyengar (2023); Uysal, Li & Mulvey (2024)	Aligns loss and decision	Small universes; stability; scale
Reinforcement learning	Cong et al. (2021); Moody & Saffell (2001)	Flexible objectives	Sample inefficiency; overfitting; replication risk
Factor timing / conditional factor weights	Haddad, Kozak & Santosh (2020); Neuhierl et al. (2024); DeMiguel, Martin-Utrera & Uppal (2024)	Captures time-varying efficacy	Factor-level; limited stock-level implementation
Regime models	Ang & Bekaert (2002); Guidolin & Timmermann (2007); Nystrup, Madsen & Lindstrom (2018); Shu, Yu & Mulvey (2025); Collin-Dufresne, Daniel & Saglam (2020)	Economic state interpretation	Few regime episodes; low-dimensional; costs rarely central
Uncertainty-aware portfolios	Barroso & Saxena (2022); Liao et al. (2025); Goldfarb & Iyengar (2003)	Temper noisy alphas	New; interaction with costs untested

2. Gap Taxonomy (status after systematic comparison)

2.1 Alpha combination

Topic	Status	Evidence	Gap statement
Combining heterogeneous signals	Solved (prediction); Partially (net)	Gu, Kelly & Xiu (2020); DeMiguel et al. (2020); Jensen et al. (2026)	Net value of heterogeneity through netting is shown linearly; nonlinear netting benefits unmeasured
Dynamic signal weighting	Partially solved	Neuhierl et al. (2024); DeMiguel, Martin-Utrera & Uppal (2024); Haddad, Kozak & Santosh (2020)	Stock-level, high-dimensional, cost-aware conditional weights untested
Nonlinear signal interactions	Solved (prediction)	Gu, Kelly & Xiu (2020); Bryzgalova, Pelger & Zhu (2025)	Net-of-cost value of interactions vs their turnover cost unmeasured
Cross-sectional vs time-series prediction	Largely solved	Gu, Kelly & Xiu (2020); Dong et al. (2022)	Low marginal value as stand-alone topic
Regime-dependent efficacy	Partially solved (descriptive)	Stambaugh, Yu & Yuan (2012); Daniel & Moskowitz (2016); Muller & Preissler (2026)	Exploiting it implementably at stock level is open
Signal decay	Solved (descriptive)	McLean & Pontiff (2016); Chen & Velikov (2023); Chen & Welch (2026)	Decay-aware combination partially addressed via horizons (Blitz et al., 2023; Jensen et al., 2026)
Signal redundancy	Solved (linear)	Green, Hand & Zhang (2017); Freyberger, Neuhierl & Weber (2020); Jensen, Kelly & Pedersen (2023)	Low novelty
Correlated alpha signals	Solved (shrinkage)	Kozak, Nagel & Santosh (2020)	Low novelty
Alpha crowding	Open but measurement-limited	(limited peer-reviewed evidence; one withdrawn arXiv paper)	Data constraints on crowding proxies
Meta-learning for alpha weighting	Open (early)	Wang & Lera (2025); Zhao, Kong & Shen (2023)	Net value untested; risk of overfitting
Uncertainty-aware combination	Partially solved	Liao et al. (2025); Barroso & Saxena (2022)	Cost interaction untested

2.2 Machine learning (value vs complexity)

Summary of the methods table in the Literature Review. Tree ensembles and NN ensembles are justified. Transformers, RL, GNNs and meta-learning are justified only as robustness or extension cells unless they survive cost-aware objectives and snooping corrections [\[inference\]](#).

2.3 Portfolio construction

Topic	Status	Gap relevant to us
Mean-variance with shrinkage	Solved	Use as default optimiser with NL shrinkage (Ledoit & Wolf, 2017)
Risk parity / min variance / max diversification / HRP	Solved (risk baselines)	Not alpha benchmarks
Black-Litterman	Solved	Possible uncertainty-blending cell
CVaR / robust / DRO	Solved (methods)	Robust budgets driven by ML uncertainty: partially open
ML-driven optimisation / end-to-end	Partially solved	Scale + realistic constraints open (Butler & Kwon, 2023; Costa & Iyengar, 2023)
Transaction-cost-aware optimisation	Solved (theory and linear/ML)	(Garleanu & Pedersen, 2013; DeMiguel et al., 2020; Jensen et al., 2026)
Turnover-constrained	Solved	Benchmark variant
Dynamic / regime-dependent trading speed	Theory solved; empirics open	(Collin-Dufresne, Daniel & Saglam, 2020) prediction untested in ML setting

3. The Real Research Gaps (ranked)

G1: Attribution of net-of-cost ML value (highest). Existing papers compare *bundles*. GKX2020 compares model classes under prediction loss. JKMP2026 compares predict-then-optimize with economic-objective learners. DMU2024 shows conditioning in low dimension. SWZ2023 compares linear and nonlinear policies. No paper we found holds the signal library, universe, optimiser, cost model and inference fixed while switching nonlinearity, conditioning, objective and uncertainty on and off in a factorial design. Without that, the field cannot say *why* a given ML pipeline adds net value [inference; re-verify].

G2: The conditioning × cost interaction. Theory (Collin-Dufresne, Daniel & Saglam, 2020; Garleanu & Pedersen, 2013) implies conditioning should be valuable when trading speed and aim portfolios adapt to states. Prediction-loss models that condition on states can instead increase turnover. Whether state dependence helps only when costs are internalised is a sharp, testable hypothesis [inference].

G3: Economic interpretability of learned state dependence. Do learned weights move the way theory predicts? Examples: momentum down-weighted after market declines in high volatility (Daniel & Moskowitz, 2016); short-leg anomalies stronger after high sentiment (Stambaugh, Yu & Yuan, 2012); faster trading in liquid states (Collin-Dufresne, Daniel & Saglam, 2020). This moves the paper from "does it work" to "why it works".

G4: Uncertainty in a cost-aware combination. Whether uncertainty shrinkage mainly works by cutting turnover (hypothesis) is untested (Liao et al., 2025).

G5: Stock-level implementation of factor-timing information. Open; lower priority because it is partly nested in G1/G2 through state variables such as factor momentum and valuation spreads.

4. Open Questions for Reviewers to Probe (and our planned answers)

Likely reviewer question	Planned answer
"Isn't this JKMP2026 with extra steps?"	JKMP is a benchmark cell; our contribution is the factorial attribution, interaction test and mechanism tests. We replicate JKMP's Portfolio-ML with their public code.
"Are gains just microcaps?"	Main universe excludes microcaps (NYSE 20th percentile); microcaps only in robustness (Hou, Xue & Zhang, 2020; Avramov, Cheng & Metzker, 2023).
"How many configurations did you try?"	Every configuration is logged; DSR, PBO, SPA and MCS computed over the full log.
"Is your cost model realistic?"	Spread estimators (Ardia, Guidotti & Kroencke, 2024; Abdi & Rinaldo, 2017; Hasbrouck, 2009) plus square-root impact; 0.5x-3x sensitivity; AUM grid; FIM2018 and PW2020 as bounds.
"Are states just data-mined?"	States fixed ex ante from theory; placebo/shuffled-state tests.
"Does it work outside the US?"	JKP developed-markets robustness.

5. Pipeline Architectures A-F: Academic and Practical Comparison

Pipeline	Description	Academic novelty today	Practical value	Main risk	Role in our design
A	Signals, normalise, ML combination, expected returns, optimiser	Low (GKX2020, AIM)	High (industry standard)	Loss/objective mismatch; turnover	Prediction-loss cells (P)
B	Meta-model learns signal weights, dynamic weights, construction	Moderate	High	Few effective time-series observations	Conditional linear cells; Idea I02
C	Regime detection, regime-specific weights, optimiser	Low-moderate	Moderate	Few regime episodes; look-ahead in regime labels	Gated conditional cells with pre-specified states
D	Ensemble, uncertainty estimation, risk-aware optimisation	Moderate (emerging)	High	Uncertainty miscalibration	Uncertainty factor (U)
E	Signals + regime + volatility + liquidity, ML, joint alpha, TC-aware optimiser	Moderate	High	Conflates conditioning with nonlinearity	Decomposed explicitly in factorial design
F	Learned interaction, differentiable optimiser, direct net objective	Moderate (JKMP/SWZ nearby)	Moderate-high	Instability; scale; overfitting	Economic-loss cells (E); differentiable variant is Idea I04

Recommendation [proposed]. Do not pick one pipeline. Nest A, B/C, D, E and F inside one factorial design so that the paper *measures* which architectural ingredient matters. That is the contribution a reviewer cannot get from existing papers.

6. Regime Dependence: What to Test

- **Candidate states (pre-specified, observable at t).** Realised market volatility, e.g. 1-month S&P 500 realised volatility or VIX from 1990. Aggregate illiquidity (Amihud, 2002) or Pastor & Stambaugh (2003) liquidity. Sentiment (Baker & Wurgler, 2006). Market drawdown/bear indicator (cumulative 24-month market return below 0, as in Daniel & Moskowitz (2016)). Credit spread (BAA-AAA) and term spread. Inflation and rate-change regimes (CPI YoY, 10y yield change). Factor momentum of each theme (Ehsani & Linnainmaa, 2022). Cross-sectional signal dispersion.
- **Parameterisations.** (i) Signal $\times$ state interactions (linear conditional). (ii) Gated mixture-of-experts with 2-4 experts. (iii) State-dependent risk aversion/trading speed in the optimiser.
- **Integrity.** States enter only with information available at t. Regime labels from smoothed HMM probabilities are forbidden (look-ahead); only filtered probabilities or observable thresholds are allowed. Placebo tests use shuffled state series.

7. Opportunity Matrix

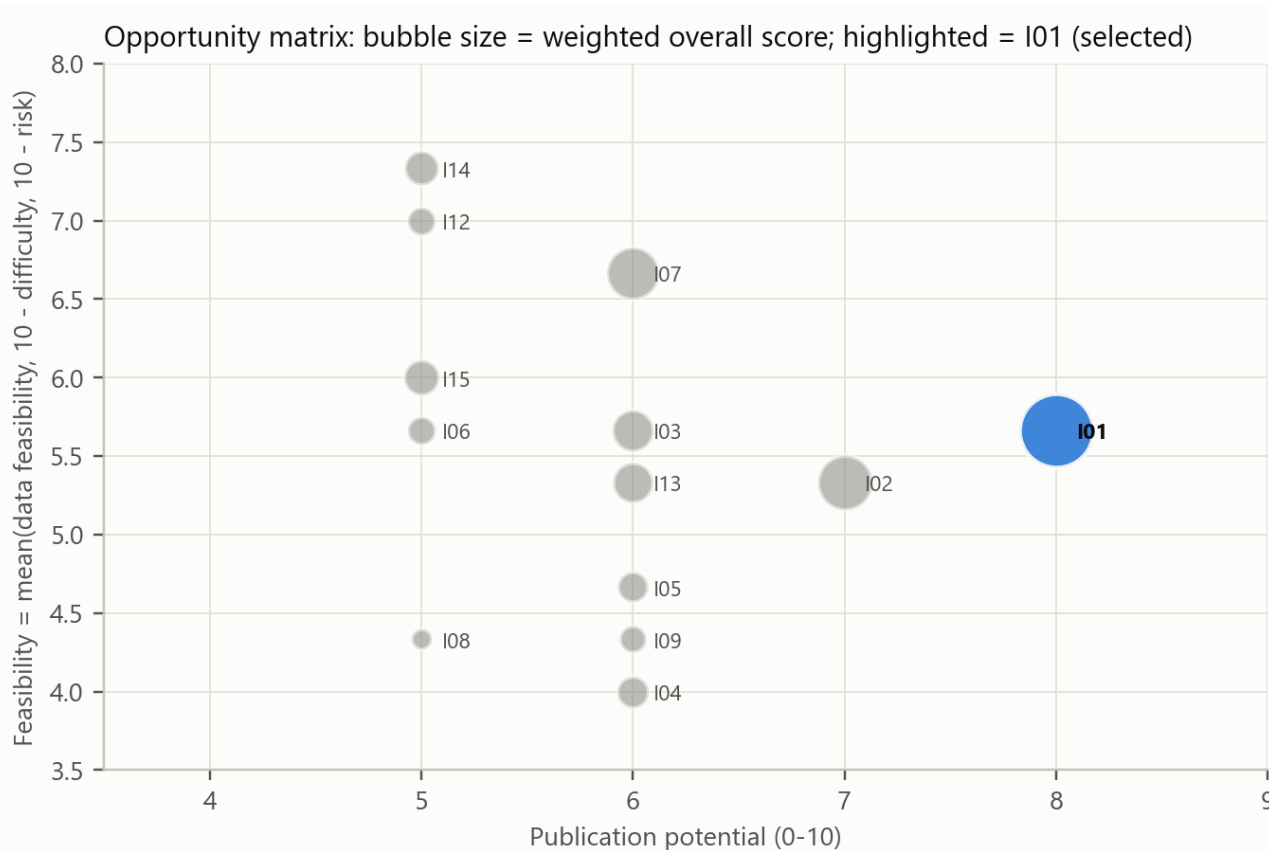


Figure 1. Opportunity matrix of the 15 candidate ideas (scores are team judgements; see Workbook 3).

Idea	Novelty	Publication potential	Data feasibility	Risk	Overall (weighted)
I01 Attribution of net ML value	8	8	8	4	Highest
I02 Signal-efficacy meta-model	7	7	8	6	High
I03 Uncertainty-aware combination	6	6	8	5	Medium-high
I04 Differentiable cost-aware layers	7	6	8	7	Medium
I05 Regime-gated MoE	6	6	8	7	Medium

(Weighted scores computed with formulas in Workbook 3; figure in PDF 3.)

Closest Prior Art Tracker (living document)

Purpose. Prevent the paper from claiming novelty that has already been published. Re-run the search protocol below at the start of every project phase, at each conference submission, and in the final week before journal submission. Record every newly found paper here, even if it seems only tangential.

Current proposed claim (Concept A). We provide a controlled, multiple-testing-robust attribution of the net-of-cost value of ML signal combination to (i) nonlinearity, (ii) state-dependent signal weights, (iii) a cost-aware economic training objective, and (iv) forecast-uncertainty shrinkage, including their interactions. We also test whether the learned state dependence matches economic theory.

Claims we must NOT make (already published):

- "First to use ML to combine many characteristics": Gu, Kelly & Xiu (2020) and many others.
- "First to incorporate transaction costs into ML portfolios": Jensen et al. (2026); DeMiguel et al. (2020) (linear); Simon, Weibels & Zimmermann (2026).
- "First to show ML alpha is fragile to costs/restrictions": Avramov, Cheng & Metzker (2023); Blitz et al. (2023); Azevedo, Hoegner & Velikov (2023).
- "First to condition on market states": Gu, Kelly & Xiu (2020) (macro interactions); Chen, Pelger & Zhu (2024) (macro states); DeMiguel, Martin-Utrera & Uppal (2024) (volatility, net of costs).
- "First to time factors with many predictors": Neuhierl et al. (2024); Haddad, Kozak & Santosh (2020).
- "First to document design-choice dispersion": Chen, Hanauer & Kalsbach (2024); Lalwani, Meshram & Jindal (2025); Menkveld et al. (2024).
- "First uncertainty-aware ML portfolio": Liao et al. (2025); Barroso & Saxena (2022).

Search protocol (repeat each phase)

1. SSRN / NBER / arXiv q-fin keyword queries: "machine learning" AND ("transaction costs" OR "implementable") AND ("signal combination" OR "characteristics"); "factor timing" AND "stock level"; "conditional" AND "parametric portfolio"; "decomposition" AND "machine learning" AND "asset pricing"; "uncertainty" AND "machine learning" AND "portfolio".
2. Forward citations (Google Scholar / Semantic Scholar / OpenAlex) of: GKX2020, JKMP2026, DMNU2020, SWZ2023, ACM2023, DMU2024, NRRZ2024, LMNS2025, CHK2024.
3. Programmes: AFA, WFA, EFA, SFS Cavalcade, NBER Asset Pricing / Big Data, SoFiE, Q-Group, INQUIRE, ICAIF.
4. Tables of contents (last 24 months): JF, RFS, JFE, JFQA, MS, RAPS, RF, QF, JFEc, JFM, JBF, JEF.

Tracker

#	Paper	Status	What it does	Overlap with our claim	Differentiation	Threat level	Last checked
1	Jensen et al. (2026)	RFS 2026	ML + trading-cost-aware learning of weights; implementable frontier; economic feature importance	Objective dimension (iii); static nonlinear economic-loss learner	We nest it as a cell; add conditioning, uncertainty, interactions, mechanism tests	High	Sep 2026
2	Simon, Weibels & Zimmermann (2026)	MS 2026 (Crossref)	Deep parametric portfolio policies; robust to costs	Nonlinear economic-objective cell	Factorial attribution; state dependence; formal snooping control	High	Sep 2026
3	DeMiguel et al. (2020)	RFS 2020	Linear PPP with TC; netting	Linear economic-loss cell	Nonlinear and conditional cells; netting attribution	Medium	Sep 2026
4	DeMiguel, Martin-Utrera & Uppal (2024)	JF 2024	Conditional multifactor (volatility) net of costs	Conditioning x costs in low dimension	Stock-level high-dimensional conditioning; many states; mechanism tests	High (for G2)	Sep 2026
5	Avramov, Cheng & Metzker (2023)	MS 2023	ML under economic restrictions	Motivation	We propose/attribute, not only diagnose	Low	Sep 2026
6	Chen, Hanauer & Kalsbach (2024)	SSRN WP	1,056 ML designs; non-standard errors	Design-grid methodology	Our factors are economic ingredients evaluated net of costs with a common optimiser	Medium	Sep 2026
7	Lalwani, Meshram & Jindal (2025)	EFM 2025	5,376 portfolios; design choices incl. portfolio construction	Design-grid methodology	Same as above	Medium	Sep 2026
8	Neuhierl et al. (2024)	SSRN WP	Factor-zoo timing	State-variable predictability	Stock-level implementation with costs	Medium	Sep 2026
9	Liao et al. (2025)	arXiv WP	NN forecast uncertainty; uncertainty-averse portfolios	Uncertainty dimension (iv)	Interaction with costs and conditioning	Medium-High (for iv)	Sep 2026
10	Azevedo, Hoegner & Velikov (2023)	SSRN WP	ML strategies net of costs and decay	Net-of-cost evaluation	Cost-aware learning, attribution	Medium	Sep 2026
11	Blitz et al. (2023)	JFDS 2023	Horizon choice and ML alpha net	Horizon/turnover lever	Horizon is a robustness factor	Low-Medium	Sep 2026
12	Cong et al. (2021)	NBER WP	RL direct construction; transformer	Direct-objective learning	Controlled attribution; costs	Low-Medium	Sep 2026
13	Collin-Dufresne, Daniel & Saglam (2020)	JFE 2020	Liquidity-regime theory with costs	Economic prediction we test	We test it empirically	Low (supportive)	Sep 2026
14	Capponi et al. (2025)	arXiv WP	Non-stationarity vs complexity	Window/complexity choice	Robustness dimension	Low	Sep 2026
15	BlackRock Systematic (2025)	Industry	AIM pipeline	Framing	Public data + inference	Low	Sep 2026
16	Zhang (2025)	arXiv WP (Oct 2025)	Friction-aware, regime-conditioned policy optimisation: costs inside the reward, inaction bands, 4 volatility-liquidity regimes, evaluated on a regime x cost grid with bootstrap CIs and multiple-testing corrections	Direct overlap with gap G2 (conditioning x cost interaction) and with our robustness grid	It proposes one RL method; we identify which of four ingredients creates net value in a ~150-signal stock cross-section, with a nested factorial design. Its regime grid becomes a robustness parameterisation we cite and reuse	High	Sep 2026

#	Paper	Status	What it does	Overlap with our claim	Differentiation	Threat level	Last checked
17	Afsharhaji & Li (2026)	arXiv WP (Apr 2026)	Sparse (basis pursuit) versus dense ridgeless selection in high-complexity feature spaces; complexity pays through sparse discovery	Our new complexity ladder (levels A2/A3)	We test the claim inside a net-of-cost, implementable setting rather than gross SDF Sharpe	Medium	Sep 2026
18	Kelly et al. (2025)	WP (Oct 2025) [verify authors/venue]	Complexity helps only when the factor eigenvalue spectrum is diffuse	Our spectrum diagnostic	We use it as a mechanism test for when the complexity rung should pay	Low-Medium	Sep 2026
19	[verify authors] (2026)	arXiv WP (2026) [verify authors]	Adaptive specification search yields significant backtests under a no-predictability null; proposes a falsification audit and multiplicity-adjusted inflation gap	Our credibility layer	We adopt it as a standard rather than compete with it; it raises the bar every rival submission must now clear	Low (supportive)	Sep 2026
20	Chen, Kelly & Malamud (2025)	arXiv WP (Dec 2025)	Limits-to-learning gap: empirical fit understates true predictability	How we interpret small R2 and IC	Cited as a caveat; no overlap of claim	Low	Sep 2026
21	Wang & Hasuiki (2026)	arXiv WP (2026)	Decision-focused (SPO) portfolio learning inflates predictions and turnover; proposes clipping, rescaling, partial adjustment	Our economic-objective cells	We adopt the controls and report inflation as a diagnostic, since the objective axis is precisely what we are testing	Medium	Sep 2026
22	Kato (2024)	arXiv WP (2024, rev. 2025)	Conformal prediction intervals for portfolio selection	Our uncertainty level D2	We use conformal as a calibrated input to a cost-aware optimiser and compare it against ensemble dispersion inside the factorial design	Medium	Sep 2026
23	Lai (2025)	arXiv WP (May 2025)	Transformer variants for US large-cap pricing; high reported out-of-sample R2	Our attention rung (A4)	Ours is a cross-sectional attention block inside a cost-aware comparison, not a standalone architecture claim	Low-Medium	Sep 2026
24	Fan et al. (2025)	arXiv WP (Dec 2024, rev. 2025)	Cost-aware mean-variance with nonconvex penalties in large universes; S&P 500 and Russell 2000	Our optimiser stage	Optimisation methodology, not attribution; we can cite it as an alternative cost-aware optimiser	Low-Medium	Sep 2026

Decision rule. If a newly found paper performs a factorial attribution across at least two of our four ingredients, net of costs, on stock-level data, pause and re-scope: shift the contribution to G2/G3 (interaction + mechanism) or to international and uncertainty extensions.

Research Ideas and Paper Concepts

1. Fifteen Candidate Research Directions

Scores (0-10) are the research team's judgement, not empirical results. Overall score = $0.20 \cdot \text{Novelty} + 0.15 \cdot \text{Academic importance} + 0.10 \cdot \text{Practical importance} + 0.20 \cdot \text{Publication potential} + 0.15 \cdot \text{Data feasibility} + 0.10 \cdot (10 - \text{Implementation difficulty}) + 0.10 \cdot (10 - \text{Failure risk})$. Workbook 3 holds all fields (research question, hypothesis, existing literature, what is solved or unsolved, exact novelty, data, signals, ML, portfolio method, benchmark, experiments, metrics, difficulties, target journals), and its weights sheet recomputes scores and ranks with live formulas.

Rank	ID	Proposed title	Nov.	Acad.	Prac.	Pub.	Data	Impl. diff.	Risk	Over all	Target journals
1	I01	Where Does Machine-Learning Alpha Come From After Trading Costs? Nonlinearity, State Dependence and Cost-Aware Learning in Signal Combination	8	9	9	8	8	7	4	7.55	RFS (stretch); JFQA, Management Science (target); Quantitative Finance, JFM, JBF (backup)
2	I02	Forecasting Signal Efficacy	7	7	9	7	8	6	6	6.75	RAPS, JFQA (target); JBF, JEF, IJF (backup)
3	I07	Netting-Aware Signal Selection	6	6	9	6	9	5	4	6.65	JFQA, FAJ, JFM
4	I03	Uncertainty-Aware Alpha Combination	6	6	8	6	8	6	5	6.20	Quantitative Finance, JFEC (target); JEF, EJOR (backup)
5	I13	State-Dependent Capacity of ML Alpha	6	6	9	6	7	6	5	6.15	JFM, FAJ
6	I15	Alpha-Risk Model Misalignment in ML Portfolios	6	5	8	5	8	5	5	5.95	QF, JPM, FAJ
7	I14	Publication-Aware Training: Eliminating Look-Ahead in Signal Discovery	4	5	7	5	9	4	3	5.90	CFR, JFM
8	I04	Differentiable Cost-Aware Portfolio Layers for High-Dimensional Equity Signal Combination	7	6	7	6	8	9	7	5.80	Management Science, Operations Research, INFORMS JOC (target); QF, EJOR (backup)
9	I05	Regime-Gated Mixture-of-Experts for Alpha Signal Combination	6	5	8	6	8	7	7	5.75	QF, JBF, JEF (target)
10	I06	Learning Signal Decay: Horizon-Aware Combination with Aim Portfolios	5	5	8	5	8	6	5	5.65	QF, JFM
11	I12	Bayesian Dynamic Model Averaging of Alpha Signals	4	4	7	5	9	4	4	5.65	IJF, JEF
12	I09	Transfer Learning of Alpha Combination Across 90+ Markets	6	6	7	6	6	7	6	5.60	JFQA, RF, JBF
13	I10	Do Text and LLM Signals Add Net-of-Cost Value to a Mature Signal Library?	7	6	8	6	4	7	7	5.50	JFQA, MS
14	I08	Crowding-Aware Alpha Combination	6	5	8	5	6	6	7	5.35	JFM, JBF
15	I11	Graph Neural Signal Propagation with Trading Costs	6	5	6	5	4	8	7	4.65	QF, EJOR

2. Five Full Paper Concepts

Concept A (from I01): "Where Does Machine-Learning Alpha Come From After Trading Costs?"

- **Research question.** Which ingredients of ML signal combination create net-of-cost investment value: nonlinearity, state-dependent weights, cost-aware economic training, forecast-uncertainty shrinkage? How do they interact?
- **Main hypothesis.** (H1) Nonlinearity raises gross IC and Sharpe, but at least half of the gain disappears net of costs. (H2) State dependence adds net value only when trading costs are internalised in training. (H3) The cost-aware objective is the largest single source of net value. (H4) Uncertainty shrinkage raises net Sharpe mainly by cutting turnover. (H5) Learned state dependence matches economic theory: momentum crashes, sentiment-driven short legs, liquidity-regime trading speed.

- **Novel contribution (proposed).** A controlled $2 \times 2 \times 2 \times 2$ factorial attribution on one public signal library, universe, optimiser and cost model, with data-snooping-robust inference and mechanism tests.
- **Literature positioning.** Builds on GKX2020 (nonlinearity), JKMP2026/DMNU2020/SWZ2023 (economic objective), DMU2024/NRRZ2024 (conditioning), LMNS2025 (uncertainty), ACM2023/BHHH2023/CV2023 (implementability), CHK2024/MENK2024 (non-standard errors).
- **Methodology.** Walk-forward annual refits, 1995-2020 development test, 2021-2025 lockbox; 16 cells; common TC-aware optimiser; factorial and Shapley attribution with block bootstrap.
- **Dataset.** CRSP, Compustat, JKP characteristics (WRDS), OpenAP signals, FF/q factors, FRED and Goyal state variables, spread estimates.
- **Signal library.** About 150 characteristics in 13 JKP themes, with publication-date flags.
- **ML architecture.** Ridge; LightGBM; NN3 deep ensemble; gated mixture-of-experts (state-only gate); economic-loss NN policy.
- **Portfolio optimizer.** Mean-variance with spread + power-law impact costs, dollar/beta/industry neutrality, ADV-based trade limits, factor risk model.
- **Benchmarks.** 1/N, market, EW/theme-EW/IC/ICIR/inverse-vol composites, Lewellen FM, ridge/ENet/PLS, RF/LightGBM/NN, KNS SDF, IPCA, DMNU PPP, JKMP Portfolio-ML, NMV banding, NRRZ-style timing.
- **Experimental design / statistical testing / robustness.** See 05 and 07.
- **Expected results (hypotheses only).** Cost-aware objective > nonlinearity in net terms; positive state $\times$ objective interaction; uncertainty reduces turnover.
- **Potential weaknesses.** Overlap perception with JKMP2026/SWZ2023; cost-model dependence; compute; null results on conditioning.
- **Why a reviewer should care.** It tells the field *what to build* and *what not to bother with* in ML portfolio construction, with inference that survives the factor zoo's multiple-testing problem.
- **Suitable journals.** RFS (stretch); JFQA, MS (primary); RAPS, RF, QF (alternatives).
- **Publication difficulty.** High (4/5).
- **What makes it publishable rather than a student project.** Pre-registered hypotheses; lockbox holdout; strongest published benchmarks run from their own code; full trial logs; cost sensitivity and capacity; mechanism tests; replication package.

Concept B (from I02): "Forecasting Signal Efficacy"

- **RQ.** Can signal-level meta-features predict each signal's next-period net contribution, and does implementing those forecasts at stock level (with netting) beat static combination?
- **Hypothesis.** Efficacy is predictable (factor momentum, valuation spread, volatility), but net benefit concentrates in low-turnover themes.
- **Novelty.** Two-level model from signal-month panel forecasts to a stock-level cost-aware portfolio; timing information evaluated after netting.
- **Positioning.** NRRZ2024, HKS2020, KNNV2024, EL2022 (timing); MP2016, CV2023 (decay); DMNU2020 (netting).
- **Methodology / data / signals.** Signal-month panel (about 150 signals $\times$ 360 months); GBDT/ridge meta-model with hierarchical shrinkage; same stock universe, optimiser and costs as Concept A.
- **Benchmarks.** Static composites; NRRZ PLS timing at factor level; factor momentum.
- **Tests.** Clark-West for efficacy forecasts; LW2008 and SPA for portfolios; placebo meta-features.
- **Weaknesses.** Few independent time-series observations; overlap with the factor-timing literature.
- **Journals.** RAPS, JFQA (target); JBF, JEF, IJF (backup). **Difficulty:** 3.5/5.

Concept C (from I03): "Uncertainty-Aware Alpha Combination"

- **RQ.** Does stock-level forecast uncertainty improve cost-aware portfolios, and through which channel: turnover, concentration or tail risk?
- **Hypothesis.** Uncertainty shrinkage cuts turnover more than gross alpha.

- **Novelty.** Uncertainty estimators (deep ensembles, bootstrap SEs, conformal) × optimiser types (shrinkage, robust ellipsoid, Black-Litterman) under costs.
- **Positioning.** LMNS2025 (closest), BS2022, GI2003, LPB2017, KZ2007.
- **Risks.** LMNS2025 could absorb much of the novelty; must emphasise the cost interaction and calibration.
- **Journals.** QF, JFEc (target); JEF, EJOR (backup). **Difficulty:** 3/5.

Concept D (from I04): "Differentiable Cost-Aware Portfolio Layers at Scale"

- **RQ.** Can end-to-end training through a constrained, cost-aware QP scale to 1,000+ stocks and beat two-stage pipelines net of costs?
- **Novelty.** Implicit differentiation with factor-structured covariance and power-law costs at stock-universe scale.
- **Positioning.** BK2023, CI2023, ULM2024, EG2022, AAB2019; JKMP2026 as economic benchmark.
- **Risks.** Compute, gradient instability, marginal gains over JKMP-style learners.
- **Journals.** MS, OR, INFORMS JOC (target); QF, EJOR (backup). **Difficulty:** 4/5.

Concept E (from I05): "Regime-Gated Mixture-of-Experts Combination"

- **RQ.** Do experts gated by observable market states beat global models net of costs, and are the gates economically interpretable?
- **Novelty.** Gates restricted to economically motivated state variables, with gate-level interpretation and placebo gates.
- **Positioning.** CPZ2024, DM2016, SY2012, CDS2020, SYM2025, JJNH1991.
- **Risks.** Few regime episodes; this concept is largely nested in Concept A's conditional cells.
- **Journals.** QF, JBF, JEF. **Difficulty:** 3/5.

3. Selection of the Best Paper Idea

Criterion	A	B	C	D	E
Novelty	8	7	6	7	6
Academic contribution	9	7	6	6	5
Practical significance	9	9	8	7	8
Data feasibility	8	8	8	8	8
Implementation feasibility	6	7	7	4	6
Statistical rigor achievable	9	7	8	7	6
Reproducibility	9	9	9	7	8
Potential performance (not required to "win")	7	7	6	6	6
Causal/mechanistic insight	8	6	6	4	6
Journal fit (finance)	9	8	7	6	6
Mean	8.2	7.5	7.1	6.2	6.5

Why Concept A is superior.

1. **Its contribution does not depend on "winning" a backtest.** Whatever the result (nonlinearity helps or not, conditioning helps or not), the attribution is informative and publishable if the design is rigorous. That lowers the failure risk that sinks most ML-finance projects.
2. **It is protected against the closest prior art.** JKMP2026, SWZ2023 and DMNU2020 become benchmark cells rather than competitors.
3. **It nests Concepts B, C and E** as factors or extensions, so the pipeline built for A supports follow-up papers.
4. **Data are obtainable** through WRDS plus open datasets.
5. **It answers "why"**, via mechanism tests tied to published theory (DM2016, SY2012, CDS2020, ACM2023).

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